Atharv Yadav

M.Sc. student, Autonomy Technologies · FAU Erlangen-Nürnberg

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Available as Werkstudent (up to 20 h/week) or intern

PROFILE

Master's student in Autonomy Technologies with a B.Tech in Electrical and Computer Engineering (German grade 1.4). I have trained and published a computer-vision model on a dataset I built myself, deployed ML models on a Raspberry Pi, and spent six months on C and Linux internals. I am moving into robotics and Physical AI: most recently I worked on a self-improving robot learning loop (ACT, SmolVLA, MuJoCo) at the Siemens Healthineers Robotics & AI Hackathon.

EDUCATION

M.Sc. Autonomy Technologies · Friedrich-Alexander-Universität Erlangen-Nürnberg — Apr 2026 – present

- Coursework: Robotics 1, Perception in Robotics, Computer Vision, Speech and Language Processing

B.Tech Electrical and Computer Engineering · Dr. Vishwanath Karad MIT World Peace University, Pune — 2021 – 2025

- CGPA 9.20 / 10 (German grade 1.4)

EXPERIENCE

Embedded Intern · eInfochips, Pune — Jan 2025 – Jul 2025

- Completed a six-month embedded systems program covering C, Linux, shell scripting and Linux internals, including the exercises in *The C Programming Language* (Kernighan and Ritchie).
- Did system-level exercises with Linux internals and command-line tools, and applied C and shell scripting in Linux-based embedded environments.

PROJECTS

AUTODIDACT · MuJoCo, ACT, SmolVLA, Python — Siemens Healthineers Robotics & AI Hackathon, Oct 2026

- Team project: a self-improving learning loop for a two-arm Franka FR3 Duo in MuJoCo. It finds where the policy fails, requests demonstrations there, retrains (ACT, SmolVLA) and only promotes a model that is statistically better.
- Contributed across data checks, the simulation twin, policy training and evaluation. The promoted model improved success by +0.20 (95% CI +0.10 to +0.30) on 72 frozen test scenarios.

3D Print Failure Detection · YOLOv8, Roboflow, Python — published, see below

- Built an image dataset of failed 3D prints from scratch, annotated it in Roboflow and trained a YOLOv8 model to tell failed prints from good ones.
- Combining our dataset with a public Roboflow dataset raised the true-positive rate on failed prints from 72% to 91% and the F1 score from 0.615 to 0.739.

MedFusion AI · TensorFlow/Keras, Random Forest, Raspberry Pi — B.Tech team project

- Diagnostic tool running separate models for COVID-19, breast cancer, brain tumor and diabetes detection, hosted on a Raspberry Pi acting as a hospital server.
- My part: dataset sourcing and preprocessing, model training and the Raspberry Pi deployment. A teammate built the web frontend.

Document RAG Pipeline · LangChain, ChromaDB, Sentence Transformers — github.com/Atharv2k4/RAG

- Question answering over text and PDF files: LangChain chunking, Sentence Transformers embeddings in ChromaDB, answers from a Groq-hosted LLM.

PUBLICATION

O. Sakharkar, **A. Yadav**, A. Agrawal and B. S. Chaudhari, “Advanced Data Fusion for Enhanced 3D Print Failure Detection”, 8th International Conference on Soft Computing and Signal Processing (ICSCSP-2025), Hyderabad, 20–21 June 2025. Proceedings in Springer Lecture Notes in Networks and Systems (LNNS). Presented the paper at the conference.

SKILLS

Programming	Python, C, Java, SQL, shell scripting
ML and vision	Computer vision (YOLOv8), TensorFlow, Keras, RAG with LangChain, ChromaDB and Sentence Transformers
Systems and embedded	Linux, Linux internals, Raspberry Pi, ESP32, edge AI
Robotics and simulation	MuJoCo, imitation learning (ACT, SmolVLA), policy evaluation, ROS 2 (fundamentals)
Automation	PLC automation with TIA Portal, Logosoft
Tools	Git, MATLAB, Roboflow, Power BI (basic)
Languages	English (fluent), German (B1), Marathi and Hindi (native)
Training	Siemens training in PLC and automation and Android development, NITTR Bhopal